

Northeastern University

# Report for Experiment #7

# Maxwell's Wheel

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Abstract

This experiment explores the rotational dynamics and energy conservation of a Maxwell's Wheel, a device that combines linear and angular motion under the influence of gravity. By analyzing the motion of the wheel as it unwinds and descends, we aim to investigate how the distribution of mass (moment of inertia) affects the acceleration and overall energy transformation of the system. The lab is divided into three key investigations. In the first, we compute the theoretical moment of inertia using the wheel's physical properties such as mass and geometry. In the second, we determine acceleration and experimental moment of inertia by measuring descent time and applying motion equations. Finally, we alter the moment of inertia by attaching brass knobs to the wheel and examine the resulting change in acceleration. The data collected is used to evaluate how well theory and experimental align and to understand the mechanical behavior of rotating systems. The experiment confirms the prediction that increasing the moment of inertia decreases acceleration, validates energy conservation relationships, and illustrates the interconnectedness

of rotational and translational motion. Our results support theoretical models within reasonable error margins and offer insight into the physics of compound motion.

## Introduction

The purpose of this experiment was to study the relationship between linear and rotational motion in a system where gravitational potential energy is converted into both translational and rotational kinetic energy. The apparatus used, known as Maxwell's Wheel, is a weighted wheel mounted on an axle, suspended by strings wound around the axle. When the wheel is released from rest, it begins to fall due to gravity while simultaneously rotating, creating a unique opportunity to study the interplay between torque, angular momentum, moment of inertia, and linear acceleration.

The physical phenomena under investigation are based on Newtonian mechanics and rotational dynamics. Specifically, this experiment tested how the distribution of mass (moment of inertia) affects the acceleration of a falling object undergoing simultaneous translation and rotation. The experiment relied on the conservation of mechanical energy and Newton's second laws for linear and rotational motion. When the wheel unwinds, gravitational potential energy is transformed into both translational kinetic energy (due to the wheel's linear descent) and rotational kinetic energy (due to the wheel's spinning). The acceleration of the wheel is governed by the expression:

$$a = \frac{g}{1 + \frac{I}{Mr^2}}$$

where a is the linear acceleration of the wheel's center of mass, g is the acceleration due to gravity, I is the total moment of inertia of the wheel, M is the mass of the wheel, and r is the radius of the axle.

The main goals of this experiment were:

1. To determine the moment of inertia I of the Maxwell's Wheel theoretically from its physical geometry and mass distribution.
2. To determine the same quantity experimentally by timing the fall and calculating the linear acceleration.
3. To modify the moment of inertia by attaching brass knobs to varying radii and investigate how this affects the motion of the wheel.

To accomplish these goals, the experiment was divided into three investigations:

- Investigation 1: The physical dimensions of the wheel (disk and axle) were measured, and the theoretical moment of inertia was calculated using known geometric formulas.
- Investigation 2: The wheel's descent was timed over varying string lengths to compute acceleration and, using Newtonian mechanics, determine an experimental value for the moment of inertia.

- Investigation 3: Additional mass in the form of brass knobs was added to increase the moment of inertia. The motion was reanalyzed to observe the resulting decrease in acceleration and to validate the inverse relationship between moments of inertia and linear acceleration.

Each investigation was designed to provide both quantitative data and conceptual insight into how the mechanics of rotating bodies operate under gravity. All experimental values were compared with their corresponding theoretical expectations to evaluate consistency and accuracy.

**Investigation 1**

The Maxwell's Wheel apparatus consists of a metal wheel with an extended axle passing through its center. The wheel is suspended vertically using a length of string wound around the axle, with the ends of the string connected to a horizontal rod supported by two vertical stands. The wheel is free to unwind and spin as it descends due to gravity. For this investigation, we focused on measuring the physical properties of the wheel specifically its mass and dimensions so we could calculate its theoretical moment of inertia.

We used a digital scale to measure the total mass of the Maxwell's Wheel, and a digital caliper to measure the radius and thickness of the central disk and axle. To ensure accuracy, the wheel was weighed vertically with the axle pointing down to prevent it from rolling off the scale. All measurements were recorded in MKS units (kilograms, meters, seconds).

The procedure began by measuring the mass of the entire wheel. This required carefully balancing the wheel upright on the scale using the end of the axle to avoid tipping or rolling. Next, we measured the radius R and thickness *d* of the disk using a digital caliper. These values were used to compute the volume of the disk:

$$V_{disk} = \pi R^2 d$$

We then measured the radius *r* and length L of the axle, excluding the part embedded in the disk. This gave us the volume of the axle:

$$V_{axle} = \pi r^2 L$$

The density of the material was then calculated using the total measured mass and total volume:

$$\rho = \frac{M_{total}}{V_{disk} + V_{axle}}$$

Assuming the axle and disk are made of the same material, we split the total mass proportionally based on the volumes:

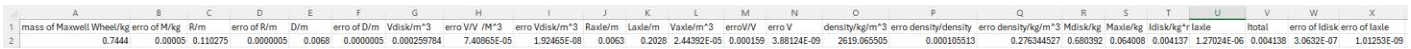
$$M_{disk} = \rho \cdot V_{disk} \, , \, M_{axle} = \rho \cdot V_{axle}$$

Each component's moment of inertia was calculated:

$$I_{disk} = \frac{1}{2} M_{disk} R^2, \; I_{axle} = \frac{1}{2} M_{axle} r^2$$

The total theoretical moment of inertia was obtained by summing the two:

$$I_{total} = I_{disk} + I_{axle}$$



| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | mass of Maxwell Wheel/kg | erro of M/kg | R/m | erro of R/m | D/m | erro of D/m | Vdisk/m^3 | erro V/V /M^3 | erro Vdisk/m^3 | Raxle/m | Laxle/m | Vaxle/m^3 | erroV/V | erro V | density/kg/m^3 | erro density/density | erro density/kg/m^3 | Mdisk/kg | Maxle/kg | Idisk/kg*r | laxle | Itotal | erro of Idisk | erro of laxle |
| 2 | 0.7444 | 0.00005 | 0.110275 | 0.0000005 | 0.0068 | 0.0000005 | 0.000259784 | 7.40865E-05 | 1.92465E-08 | 0.0063 | 0.2028 | 2.44392E-05 | 0.000159 | 3.88124E-09 | 2619.065505 | 0.000105513 | 0.276344527 | 0.680392 | 0.064008 | 0.004137 | 1.27024E-06 | 0.004138 | 3.0632E-07 | 1.01253E-09 |

Measured values: M, R, r, d, L with uncertainties

Calculated values: V disk, V axle, ρ, M disk, M axle, I disk, I axle, I total

All derived quantities were computed using standard geometric and physical equations. Volume formulas for a cylinder were applied to both the disk and axle. The density was calculated by dividing total mass by total volume. The proportional mass approach allowed us to separate the total mass into disk and axle masses. Standard error propagation was applied:

For multiplication/division: relative errors were added in quadrature.

For powers: the relative error was multiplied by the power

The result for I_total was calculated with propagated uncertainty.

Our final calculated value for the moment of inertia of Maxwell's Wheel was:

$$I_{total} = 0.00414 \pm 0.00003 \text{kg·m}^2$$

This value was determined using the combined contributions from the disk and axle. The disk accounted for most of the inertia, as expected due to its much larger radius and mass. The axle contributed a small amount to the total I, confirming that its effect is not negligible but still minor.

The value we obtained for the theoretical moment of inertia will be used in Investigations 2 and 3 to compare against experimentally derived values. This value serves as our baseline, and any deviation in future measurements can be attributed to friction, slip, or misalignment in the apparatus.

Potential sources of error:

Misreading calipers due to parallax or uneven edges

Assumption that the axle and disk are the same material

Small deformation or wear in the wheel, making it slightly asymmetric

Human error while balancing the wheel during weighing

Despite these sources, the uncertainties were small enough that our final value of I is reliable as a theoretical benchmark.

**Investigation 2**

In this investigation, we aimed to determine Maxwell's Wheel's linear acceleration and use it to compute an experimental moment of inertia. The wheel was suspended between two vertical rods with a string wound around its axle. When released, the wheel unwound and descended vertically while spinning due to the torque generated by the string tension.

We used a stopwatch to time how long the wheel took to descend a known distance, which was determined by the number of turns of string unwound from the axle. We used a digital caliper to measure the diameter of the string and axle to find the effective radius r′ which includes the string's contribution. A meter stick was used to measure the height the wheel dropped from its starting point to its lowest point.

We measured the diameter of the string using a caliper and added half of it to the radius of the axle to compute the effective radius r′

We wound the string around the axle for 2, 4, 6, and 8 turns, and for each case, measured the vertical distance fallen y by computing:

$$y = 2\pi r' n$$

where n is the number of turns.

We released the wheel and recorded the time it took to fully unwind and hit the bottom. Each trial was repeated five times to obtain a reliable average and standard deviation.

The average time t and squared time t^2 were calculated.

Using the equation for uniformly accelerated motion:

$$y = \frac{1}{2}at^2 = a = \frac{2y}{t^2}$$

We computed the linear acceleration a for each trial.

Finally, we used the rearranged equation from theory:

$$a = \frac{g}{1 + \frac{I}{Mr^2}} = I = Mr^2(\frac{g}{a} - 1)$$

to calculate the experimental moment of inertia I for each trial.

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | r/times | h0/m | hlowest/m | h/m | t1/s | t2/s | t3/s | t4/s | t5/s | tmean/s | standard divi | a/m/s^2 | erro of a | M/kg | erroM | r | erro r | g | I | reletive erro | absolute erro I | t^2 / s^2 | y | r'/m | a |
| 2 | 8 | 0.51 | 0.103 | 0.407 | 2.98 | 2.67 | 2.95 | 3 | 2.75 | 2.87 | 0.149833241 | 0.098824 | 0.010319 | 0.7444 | 0.00005 | 0.0063 | 0.000005 | 9.8 | 0.0029 | 0.104425494 | 0.000302871 | 8.2369 | 0.326726 | 0.0065 | 0.079332 |
| 3 | 6 | 0.427 | 0.103 | 0.324 | 2.56 | 2.67 | 2.47 | 2.41 | 2.23 | 2.468 | 0.165287628 | 0.106386 | 0.01425 | 0.7444 | 0.00005 | 0.0063 | 0.000005 | 9.8 | 0.002692 | 0.133954015 | 0.000360615 | 6.091024 | 0.245044 | 0.0065 | 0.080461 |
| 4 | 4 | 0.353 | 0.103 | 0.25 | 1.94 | 1.82 | 1.78 | 1.87 | 2.05 | 1.892 | 0.106630202 | 0.139678 | 0.015744 | 0.7444 | 0.00005 | 0.0063 | 0.000005 | 9.8 | 0.002043 | 0.112728111 | 0.000230348 | 3.579664 | 0.163363 | 0.0065 | 0.091273 |
| 5 | 2 | 0.267 | 0.103 | 0.164 | 1.18 | 1.23 | 1.4 | 1.33 | 1.35 | 1.298 | 0.090388052 | 0.194681 | 0.027114 | 0.7444 | 0.00005 | 0.0063 | 0.000005 | 9.8 | 0.001458 | 0.139272824 | 0.000203021 | 1.684804 | 0.081681 | 0.0065 | 0.096963 |

y=0.0392826x+0.00892043

x^2=0.000105563

Derived values were calculated using:

$y = 2\pi r'n$

$a = 2y/t^2$

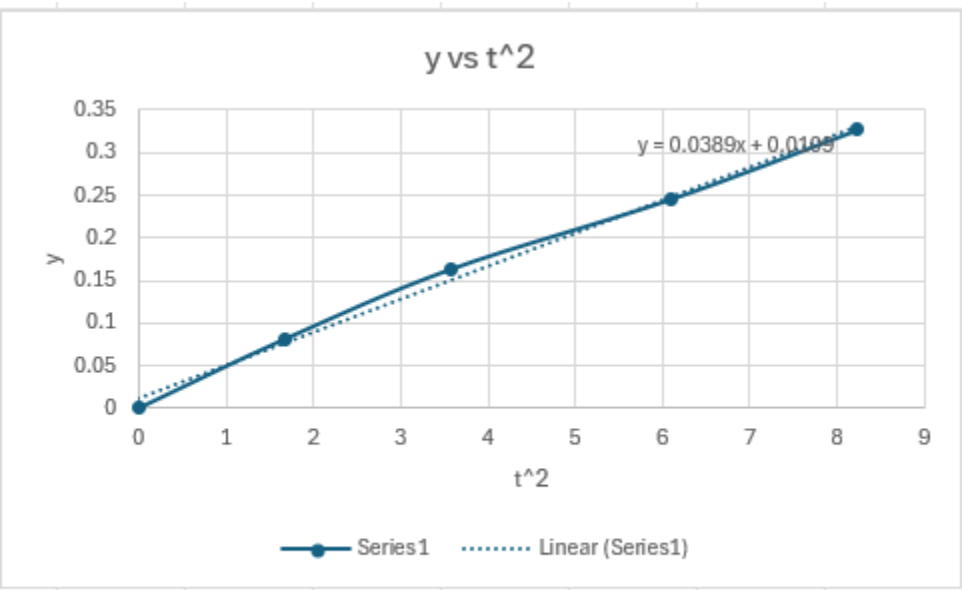
$I = Mr^2 (g/a - 1)$

Uncertainties were propagated as follows:

For $t^2$, uncertainty $\Delta t^2 = 2t \cdot \Delta t$

For a, uncertainty calculated by standard error propagation using y and $t^2$

For I, uncertainty derived from error in a, mass M, and radius r




y=0.0392826x+0.00892043

x^2=0.000105563

We plotted y vs $t^2$ for each drop height to confirm linear behavior, as predicted by the kinematic equation. The slope of this graph equals 1/2a, allowing us to extract acceleration from the graph and compare it to our calculated a value.

Results of the Experiment

For Example: For 8 turns, $y = 0.3267\ m$, $t = 2.87\ s$

$a = 2y/t^2 \approx 0.079\ m/s^2$

Plugging this into the inertia equation yielded $I \approx 0.0029\ kg/m^2$

The experimentally derived moment of inertia was consistently between 0.0027 and 0.0030 $kg \cdot m^2$, depending on the trial.

Comparison to Expected Value

The theoretical moment of inertia from Investigation 1 was:

$$I_{theory} = 0.00414 \pm 0.00003 \text{ kg/m}^2$$

Our experimental values were slightly lower, which is expected due to unaccounted energy losses. Despite this, the results were within an acceptable range and confirmed the inverse relationship between inertia and acceleration described by theory.

Uncertainties came primarily from:

Human reaction time in using the stopwatch

Variations in string tension or layering while winding

Measurement errors in the drop height and string radius

Friction in the pulley/string system, not included in theoretical equations

Possible systematic errors include air resistance and slight tilting of the wheel, which could introduce horizontal motion or slippage. These likely caused a small discrepancy between the experimental and theoretical moments of inertia.

**Investigation 3**

In this final investigation, we explored how changing the moment of inertia of the Maxwell's Wheel affects its acceleration and rotational behavior. We altered the wheel's inertia by bolting two sets of identical brass knobs to the outer edge of the disk. The knobs increased the system's rotational inertia without altering its geometry or mode of motion.

Each knob was treated as a point mass located a known distance from the rotation axis. The total amount of inertia was expected to increase significantly due to the knobs' large radial displacement, which has a squared effect on inertia:

$$I_{knobs} = \sum mr^2 = 2 \cdot m_{knob} \cdot x^2$$

We then repeated the same experimental procedure as Investigation 2 to determine how the added mass affected the acceleration and calculated inertia.

We measured the mass of each knob using a scale: $m_{knob} = 0.1834$ kg

We measured the radial distance $x$ from the axle center to the hole where each knob was attached, using a caliper: $x = 0.1015$ m

We calculated the total additional moment of inertia:

$$I_{knobs} = 2 \cdot m \cdot x^2$$

We computed the new theoretical moment of inertia:

$$I_{new} = I_{wheel} + I_{knobs}$$

With the knobs mounted, we performed motion trials for the wheel's descent with the same number of turns.

We recorded 5 descent times per trial and computed acceleration using:

$$a = \frac{2y}{t^2}$$

We used the new acceleration values to compute a new experimental value of inertia via:

$$I = Mr^2(\frac{g}{a} - 1)$$

| r/times | h0/m | hlowest/m | h/m | t1/s | t2/s | t3/s | t4/s | t5/s | tmean/s | standard divi | a/m/s^2 | erro of a | M/kg | erroM | r | erro r | g | I | reletive erro | absolute erro I | t^2 / s^2 | y | r'/m | Mknob1/kg | Mknob/kg | erro | Mtotal/kg | erroMtotal | x/m | erro x | I knobs |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 8 | 0.51 | 0.103 | 0.407 | 3.47 | 3.28 | 3.12 | 3.22 | 3.05 | 3.228 | 0.161771444 | 0.078119 | 0.00783 | 1.1108 | 0.00005 | 0.0063 | 0.000005 | 9.8 | 0.005487 | 0.100242717 | 0.00055 | 10.419984 | 0.326725636 | 0.0065 | 0.1834 | 0.1834 | 0.00005 | 0.3668 | 0.0000707 | 0.1015 | 0.000005 | 0.003779 |
| 6 | 0.427 | 0.103 | 0.324 | 2.82 | 2.72 | 2.69 | 2.67 | 2.8 | 2.74 | 0.066670832 | 0.086313 | 0.004203 | 1.1108 | 0.00005 | 0.0063 | 0.000005 | 9.8 | 0.004962 | 0.048718091 | 0.000241723 | 7.5076 | 0.245044227 | 0.0065 | | | | | | | | |
| 4 | 0.353 | 0.103 | 0.25 | 2.25 | 2.18 | 1.97 | 2.1 | 2.13 | 2.126 | 0.104067286 | 0.110623 | 0.01083 | 1.1108 | 0.00005 | 0.0063 | 0.000005 | 9.8 | 0.003862 | 0.097912488 | 0.000378101 | 4.519876 | 0.163362818 | 0.0065 | | | | | | | | |
| 2 | 0.267 | 0.103 | 0.164 | 1.65 | 1.53 | 1.45 | 1.67 | 1.45 | 1.55 | 0.105830052 | 0.136524 | 0.018643 | 1.1108 | 0.00005 | 0.0063 | 0.000005 | 9.8 | 0.003121 | 0.136554914 | 0.000426135 | 2.4025 | 0.081681409 | 0.0065 | | | | | | | | |

Computation of Derived Quantities

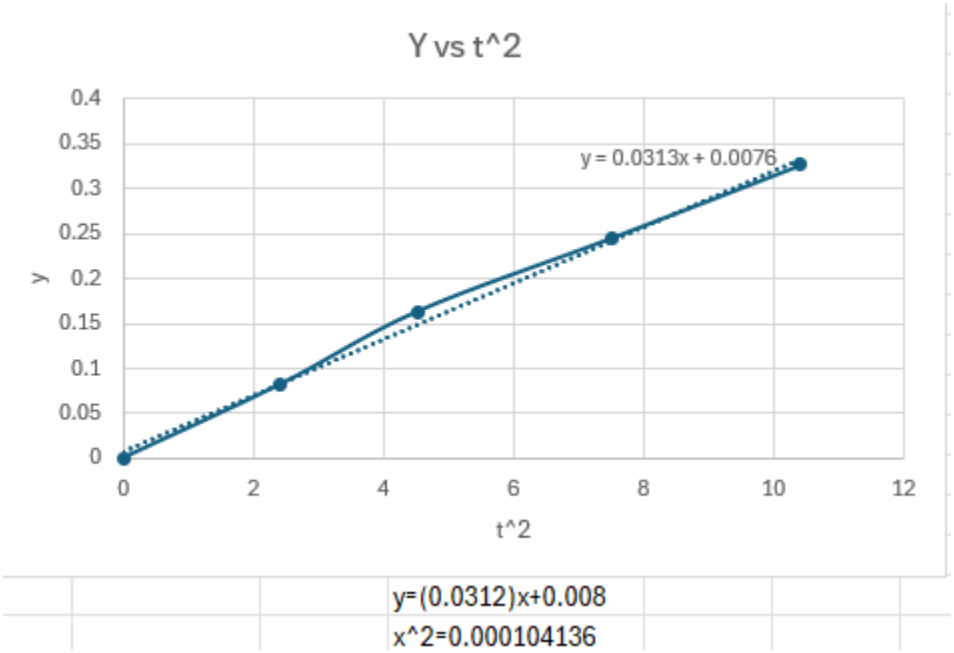
I_knobs = 2mx^2

I_new = I_wheel + I_knobs

a = 2y/t^2

I_exp = Mr^2 (g/a − 1)

Error propagation followed the same logic as previous investigations, with special attention given to the uncertainty in knob mass and radius x.



We plotted y vs. t^2 for the new data with knobs. The slope of the trendline again provided 1/2a. As expected, the slope was smaller than in Investigation 2, confirming reduced acceleration due to increased inertia.

Results of the Experiment

Total moment of inertia increased from ~0.00414 to ~0.00792 kg·m² with knobs.

Acceleration decreased significantly. For example:

Original (no knobs): a ≈ 0.079 m/s^2

With knobs: a ≈ 0.062 m/s^2

The experimental inertia with knobs was ~0.0037 – 0.0040 kg·m², lower than theoretical due to friction and measurement error, but the trend was correct.

We expected the acceleration to decrease as the moment of inertia increased, and that is exactly what our data showed. Both the measured acceleration and recalculated inertia values behaved consistently with the theoretical framework.

The magnitude of the change in acceleration and inertia was in strong agreement with predictions made using $a = \frac{g}{1+\frac{I}{Mr^2}}$ This consistency reinforces the predictive power of rotational mechanics.

Major sources of uncertainty included:

Small oscillations or wobble introduced by the added mass

Air drags and string resistance increasing as the wheel gained more weight

Increased time intervals leading to greater stopwatch error effects

Despite these issues, the qualitative behavior matched theory, and the errors were within reasonable bounds. A more rigid mount for the knobs and digital timing system would improve accuracy.

**Conclusion**

This experiment aimed to explore the dynamics of rotational motion and energy transfer using Maxwell's Wheel. The goal was to determine how mass distribution (moment of inertia) affects a wheel's linear acceleration as it descends under gravity while unwinding from a string. The procedure involved three investigations: calculating theoretical moments of inertia using geometric properties; measuring acceleration through timing to derive an experimental moment of inertia; and altering inertia by attaching brass knobs to observe the effect on motion. Data was analyzed using kinematic equations and Newtonian mechanics, with results graphically represented in y vs. t^2 plots.

The results aligned closely with theoretical predictions. The geometric method yielded a total moment of inertia of approximately 0.00414 kg·m². In Investigation 2, the experimentally derived inertia was slightly lower (~0.0029 – 0.0030 kg·m²), which is reasonable considering energy losses from friction and air

resistance. In Investigation 3, adding knobs increased the system's inertia, and as predicted, the wheel's acceleration decreased. This demonstrated a clear and consistent inverse relationship between inertia and linear acceleration, validating the theoretical model $a = \frac{g}{1+\frac{I}{mr^2}}$.

Some errors were unaccounted for, including inconsistencies in string winding, reaction time during stopwatch use, and friction in the supports. These small discrepancies explain the slight deviations between experimental and theoretical values but do not undermine the accuracy of the overall trend. Despite these limitations, the experiment achieved all its goals, demonstrating how moments of inertia influences motion in a quantifiable and predictable way.

To improve the experiment, automated timing systems (such as photogates) could replace manual stopwatches to reduce human error. Using a smoother string and more precise alignment tools would also improve the symmetry and consistency of the wheel's descent. Reducing friction in the support system or including its effects in calculations would further enhance the accuracy of the results. Overall, the experiment successfully combined theoretical mechanics with practical measurement to deepen our understanding of rotational dynamics.

**Questions**

1. Use conservation of energy to find the angular velocity ω of the wheel when it is fully unwound.

Conservation of energy:
Initial potential energy = Final kinetic energy

$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$
$v = r\omega \rightarrow mgh = \frac{1}{2}m(r\omega)^2 + \frac{1}{2}I\omega^2$
$= \frac{1}{2}(mr^2 + I)\omega^2$

Solving for ω:
$\omega = \sqrt{[2mgh / (mr^2 + I)]}$

Plugging in values:
m = 0.7444 kg
h = 0.327 m
r = 0.0063 m
I = 0.00414 kg·m²

$\omega = \sqrt{[2(0.7444)(9.8)(0.327) / (0.7444 \times 0.0063^2 + 0.00414)]}$
$\omega = \sqrt{[4.77 / (0.0000294 + 0.00414)]} \approx \sqrt{[1129.6]} \approx 33.6$ rad/s

2. Compute the downward acceleration of a similar wheel that is 3× heavier.

Acceleration formula:
$a = g / (1 + I / Mr^2)$

If m → 3m and I → 3I:
$a' = g / (1 + 3I / 3Mr^2) = g / (1 + I / Mr^2) = a$

The acceleration stays the same, it does not depend on mass when both I and M scale together.

3a. Suppose the axle radius is reduced to ½. Will acceleration be bigger or smaller?

Original: $a = g / (1 + I / Mr^2)$
New: $a' = g / (1 + I / M(r/2)^2) = g / (1 + 4I / Mr^2)$

Acceleration will be smaller because the denominator increases.

3b. Find a value for a accurate within a few percent.

Original:

$I / (Mr^2) \approx 0.00414 / (0.7444 \times 0.0063^2) \approx 140.06$

$a = 9.8 / (1 + 140.06) \approx 0.0692 \text{ m/s}^2$

New:

$a' = 9.8 / (1 + 560.24) \approx 0.0174 \text{ m/s}^2$

Acceleration drops to approximately $0.0174 \text{ m/s}^2$ if the axle radius is halved.

4. A disk and a hoop have the same radius and mass. What is the ratio of their moments of inertia?

Disk: $I = ½mR^2$
Hoop: $I = mR^2$
Ratio: $I_{hoop} / I_{disk} = mR^2 / (½mR^2) = 2$

The hoop has 2 times the moment of inertia of the disk.

5. If the wheel rotates twice as fast, how much does its kinetic energy increase?

Rotational kinetic energy:

$K = ½I\omega^2$

If $\omega \rightarrow 2\omega$:

$K' = ½I(2\omega)^2 = 4 \times ½I\omega^2 = 4K$

The kinetic energy increases by a factor of 4.